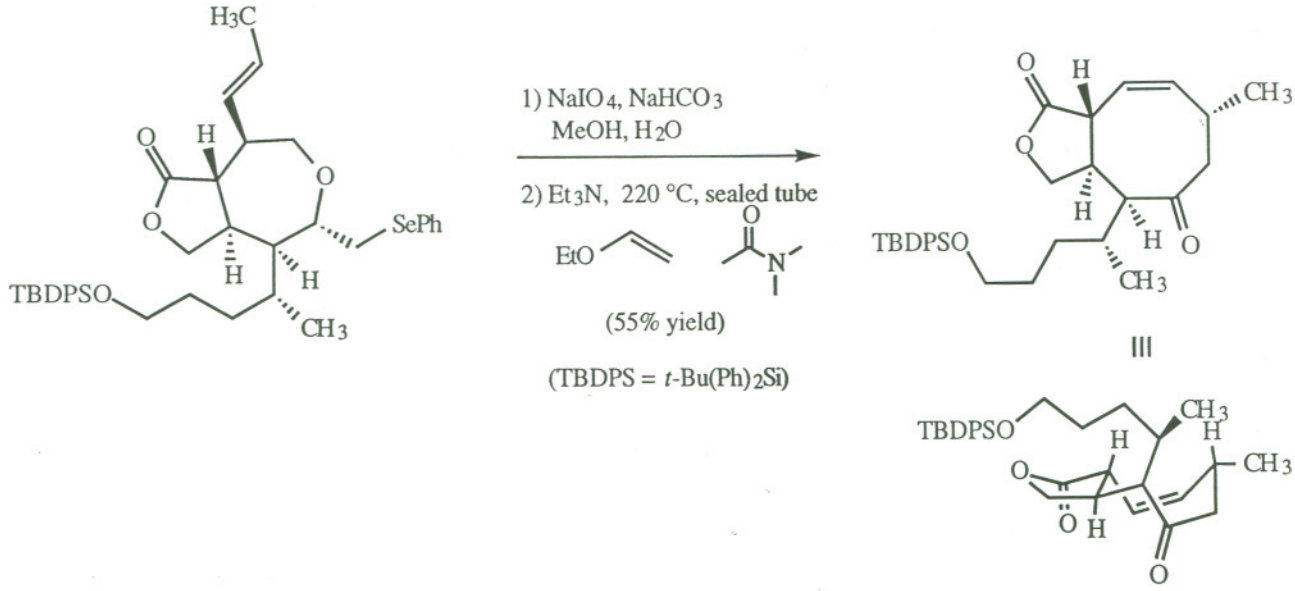
A recent article in *J. Am. Chem. Soc.* (Wang, Pinard & Paquette, *JACS* **1996**, *118*, 1309-18) reported on the "Asymmetric Synthesis of the Diterpenoid Marine Toxin (+)-Acetoxycrenulide."

The following two step process was a key element in their synthetic strategy:



The second step involved two tandem reactions (two reactions occurred consecutively in a single pot under the conditions) which required a great deal of optimization in order to accomplish the desired conversion. The role of the individual reagents required were as follows:

(i) Triethylamine was added to neutralize acidic by-products formed in the conversion.

(ii) Ethylvinylether was added as a reactive functional group related to a functional group on the intermediate compound formed. This was added so that it would selectively undergo a side reaction that was occuring on the intermediate and thus causing low yields of the desired product. The ethylvinylether was presumably more reactive and available in order to ward off attack of the related functional group in the intermediate.

(iii) N,N-Dimethylacetamide was the solvent used in place of DMF (N,N-dimethylformamide).

Provide the following in order to explain the formation of the final product:

(1) The product formed after the first reaction with $NaIO_4$.

(2) The first intermediate formed in the second reaction including the mechanism of its formation.

(3) The mechanism of the second step of the second reaction. Show the mechanism conformationally in order to rationalize the stereoselectivity of the reaction (the product is drawn conformationally to assist you).